Mechanics of Flexible Structures & Soft Robots
MAE-263F
Fall 2026

Project Proposal

Due Date: 11:59 PM, 11/09/2026

You should have a single GitHub repository for this class, shared with the instructor (`khalidjm@seas.ucla.edu`). All the homeworks, reports, and the proposal should be uploaded to this repository. Do not create a separate repository for each assignment. Within your repository, create a separate folder for each assignment (e.g., `homework_1`, `homework_2`, `homework_3`, `homework_4`, `proposal`, `midterm_report`, and `final_report`).

Submission Instructions

Your submission on BruinLearn should **only** contain the URL to the `proposal` folder of your GitHub repository, for example:

`https://github.com/<username>/<repository>/tree/main/proposal`

(Open the `proposal` folder on GitHub and copy the URL from your browser. If your default branch is not `main`, the URL will show its name instead.)

The `proposal` folder should contain:

1. **Project proposal** in `.pdf` format. The file name should be `Proposal.pdf`.

Make sure the instructor can open the link. If your repository is private, it must be shared with `khalidjm@seas.ucla.edu`; otherwise the proposal cannot be graded.

Teams

- You may work individually or in a team of **up to 2 students**.
- A team needs only **one** GitHub repository for the project. Both team members should have access to it, and it must be shared with the instructor.
- **Each student must submit the URL on BruinLearn**, even if both team members submit the same URL.
- List the names of all team members on the first page of the proposal.

Formatting Requirements

These follow the syllabus.

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- Use one of the following two templates:
  - LaTeX template (highly recommended): `http://ras.papercept.net/conferences/support/tex.php`
  - MS Word template: `http://ras.papercept.net/conferences/support/word.php`
- **Maximum 4 pages**, excluding the bibliography.
- At least **5 references**. These must be journal articles or conference proceedings published in recognized venues.
- Include a bibliography in an acceptable format (e.g., APA).

Suggested Content

The proposal should describe what you plan to do for your course project and how you will do it. The structure below is a **suggestion only; it is not required**. You may organize your proposal differently.

1. **Title and team members.**
2. **Introduction and motivation:** the problem you want to study and why it matters (e.g., an application in flexible structures or soft robotics). We suggest using your simulation to solve a challenge, e.g., design, optimization, control, or learning.
3. **Related work:** what has been done before, supported by your references.
4. **Model of the physical system:** a schematic or drawing of the model, with a description of its degrees of freedom, forces, energies, and boundary conditions (similar to the mass-spring-damper system in Fig. 2.1 of the course notes).
5. **Algorithms:** at least one simulation algorithm, written as pseudocode (similar to Algorithm 1, Discrete Elastic Rods, in the course notes), along with any other algorithm you plan to use (e.g., for optimization, control, or learning).
6. **Expected results and validation:** what you plan to show, and how you will check that your results are correct (e.g., comparison with theory, experiments, or published results).
7. **References.**

Generative AI

As stated in the syllabus, you may use large language models (LLMs) and other generative AI tools to help write code for the course project. However, you are responsible for every line of code you submit and must be able to explain it when asked in person.

Academic integrity: you are responsible for everything in your proposal. Citing references that do not exist or do not support your statements (e.g., citations "hallucinated" by an AI tool), or including code, equations, or other content that you cannot explain when asked in person, will be treated as a violation of academic integrity. Check every reference yourself.

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